Sora Kim

HCI Researcher · AI-Supported Reasoning and Accountability in Regulated Systems · Seoul, South Korea

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RESEARCH FOCUS

> I study how accountability is structured in regulated work — and how system design shapes which causal explanations remain available before responsibility settles. My research combines HCI experimentation with STS analysis, using pharmaceutical GMP environments as a primary empirical site. Intervention work deploys AI not to accelerate conclusions but to resist premature closure, preserving the comparability of alternative explanations before attribution stabilizes.

Interests: HCI · AI-supported reasoning · accountability · regulated organizations · document infrastructure · sociotechnical systems · STS

RESEARCH PROGRAM · 9 PAPERS · PORTFOLIO

- HCI **How Regulated Systems Produce Responsibility: A Four-Layer Framework for Interaction Design**
  Umbrella paper · SSRN (forthcoming) · 2026
  Introduces Regulatory UX as a framework synthesizing four empirical studies into a unified account of how system design shapes accountability conditions.
- HCI **Keeping Alternatives Alive: A Comparison-Preserving Scaffold for Root Cause Analysis Under Closure Pressure**
  Flagship · SSRN (forthcoming) · 2026 · N=20 + N=20 **CSCW 2026 poster**
  Two-study program. AI scaffold as comparison enforcer: 18/20 participants broadened causal reasoning, 15/20 reconsidered initial attribution after scaffolded comparison.
- HCI **From Retrieval to Reasoning: How SOP Accessibility Shapes Interpretation in Regulated Work**
  SSRN (forthcoming) · 2026 · N=20
  Retrieval layer over 500 synthetic GMP SOPs. Reduced retrieval time ∼6 min → ∼1 min; surfaced invisible onboarding labor upstream of formal quality records.
- HCI **When Documents Don't Connect: Traceability Burden and Loopback Behavior in GMP Document Navigation**
  SSRN (forthcoming) · 2026 · N=20
  Two-study paper. Graph-based traceability visibility + behavioral loopback tracking. 30–60% reduction in path reconstruction time.
- HCI **Friction Before Failure: Making Review Burden Visible in Regulated Document Work**
  SSRN (forthcoming) · 2026 · N=20
  Six-tab dashboard + TF-IDF pattern detection over 100 synthetic deviation-CAPA records. Surfaces review burden invisible to existing quality metrics.

STS-grounded theoretical foundations:

- STS Kim, S. (2026). *Anticipated Accountability Convergence: Structuring Responsibility Before Action*. SSRN (forthcoming).
- STS Kim, S. (2026). *Structuring Judgment Under Constraint: Cognitive Friction in Procedural Texts*. SSRN (forthcoming).
- STS Kim, S. (2026). *Responsibility Convergence in Organizational Investigations*. SSRN (forthcoming).
- STS Kim, S. (2026). *Premature Accountability Convergence in Regulated Organizations*. SSRN (forthcoming). **4S Toronto 2026**

FDA Warning Letter corpus (N=50, 160 violations, 2016–2025) + two-study investigation program (N=20 + N=20).

CONFERENCE ACTIVITY

- **CSCW 2026** Poster submission — *Keeping Alternatives Alive: A Comparison-Preserving Scaffold for Root Cause Analysis Under Closure Pressure*
- **4S Toronto 2026** Abstract submission · Panel 74: TechnoPower — *Premature Accountability Convergence: Closure Pressure, Regulatory Documentation, and Counter-Convergence Design*

PROFESSIONAL EXPERIENCE

Quality Control / Quality Systems Project Lead Sep 2022–Present
Prestige Biologics · South Korea

- Led cross-functional quality system operations across validation, deviation, CAPA (Corrective Action and Preventive Action), and RCA in GMP (Good Manufacturing Practice)-regulated pharmaceutical manufacturing.
- Supported regulatory inspections and global audits, developing expertise in how documentation architecture shapes traceability, coordination, and responsibility attribution.
- Operated digital quality systems (LIMS, ELN, Veeva Vault) as everyday research sites, observing how system design produces accountability patterns that motivated the current research program.

SKILLS

Research Methods	**Technical**	**Languages**
Usability testing	Python · Streamlit	Korean (native)
Think-aloud protocols	R · GitHub	English (fluent)
Qualitative coding	LaTeX · HTML/CSS	Japanese (fluent)
Document & corpus analysis	LIMS · ELN · Veeva Vault	